Day 6 afternoon — Simpson's paradox and the workflow

Bayesian statistics for PhD researchers · Lecturer: Lu Cheng

Tuesday 17 November 2026 · Niels Jernes Vej 8A, room 0.02 (group work also in 0.01)

Marginal versus conditional comparisons · how the data arose · a fair overall comparison · the modelling workflow · the final assignment

Today

08:45	**Lecture A**	Yes/no outcomes · logistic regression · priors on log-odds · predictions
09:45	**Exercise A**	Hit or miss: logistic model in Stan · predictions · check · question set
11:10	**Peer review A**	Partner interviews · plenary · everyone back at 11:10
11:55	**Lunch**	
12:30	**Lecture B**	Simpson's paradox · the workflow · the final assignment
13:30	**Exercise B**	Does feedback help? A pilot study with a paradox · question set
15:00	**Peer review B**	Partner interviews · plenary · everyone back at 15:00
15:45	**Daily diary**	Mentimeter: 3 things learned, 3 not yet familiar

Follow-up from the morning plenary

Open points from Peer review A

Fill in during lunch: the one or two questions the plenary did not settle.

1. …
2. …

By the end of this lecture you can

1. **Recognise** Simpson's paradox: an association that reverses within groups
2. **Separate** a marginal from a conditional comparison
3. **Choose** the comparison that answers your question, from how the data arose
4. **Use** the whole modelling workflow on a problem of your own
5. **Know** what the final assignment asks and how the course is passed

Does feedback help? A pilot study

target	without feedback	with feedback
5 s	125 / 160 (78%)	17 / 20 (85%)
10 s	37 / 60 (62%)	42 / 60 (70%)
15 s	4 / 20 (20%)	63 / 160 (39%)
all targets	**166 / 240 (69%)**	**122 / 240 (51%)**

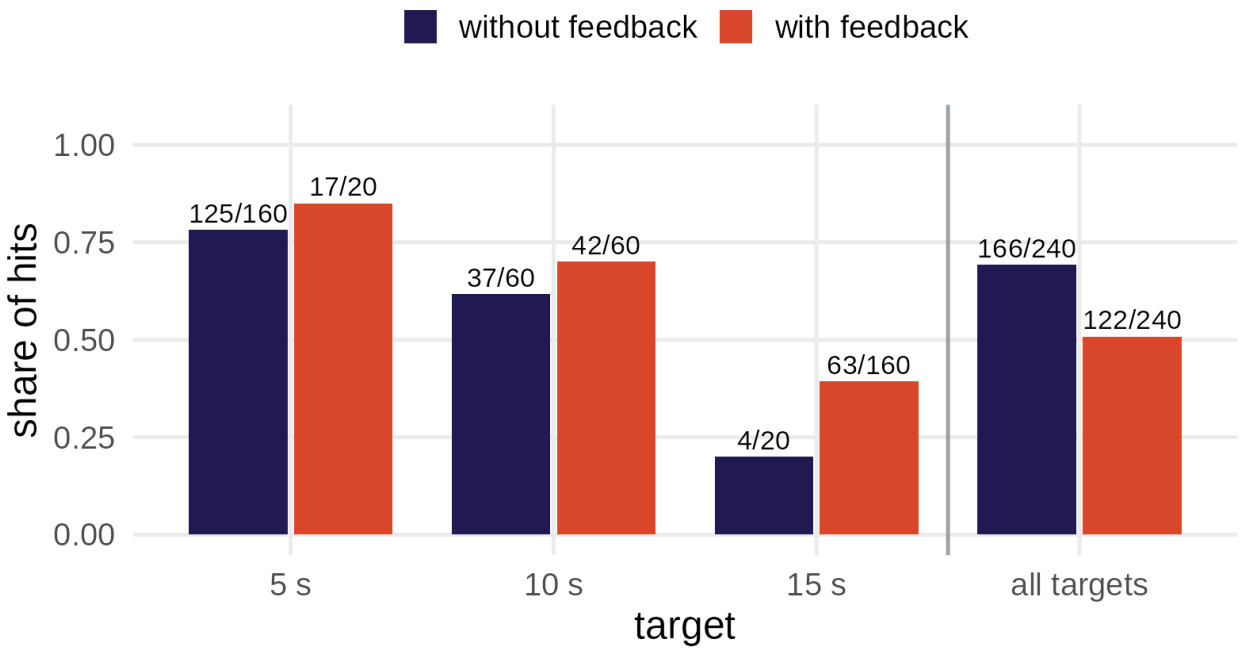
Hits out of attempts. In a pilot of the feedback game, targets were not assigned evenly: most attempts with feedback were at 15 s, most without feedback at 5 s (illustrative numbers).

Question

Does feedback help or hurt?

Read the table row by row, then the last row.

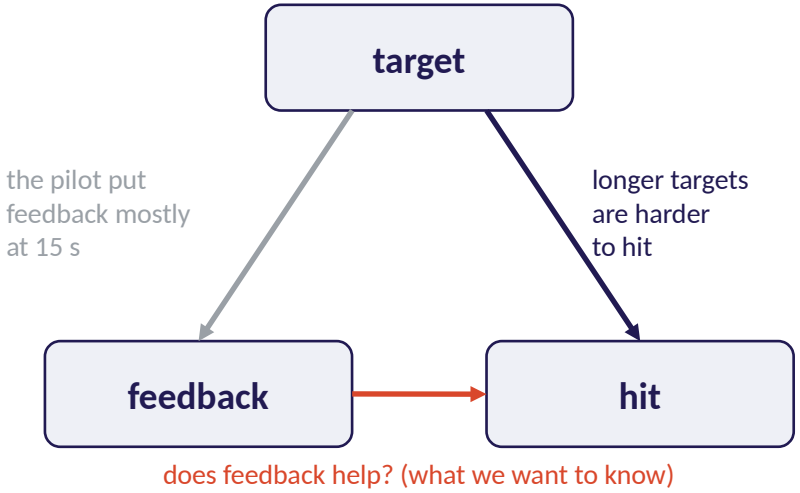
Within every target versus overall



Feedback wins at every target, and loses overall: Simpson's paradox.

Why it happens

- Hits get rarer as the target grows.
- Most attempts with feedback were at 15 s, the hardest target; most without feedback at 5 s, the easiest.
- Each overall rate is a weighted average of the three target rates — with very different weights:
  - without feedback: 160 of 240 attempts at 5 s
  - with feedback: 160 of 240 attempts at 15 s
- The overall comparison mixes the effect of feedback with the effect of the target.



Arrow: “influences”. The target influences both which condition an attempt is in and whether it is a hit, so it mixes into the overall comparison.

It happens in real data: kidney stones

	treatment A	treatment B
small stones	81 / 87 (93%)	234 / 270 (87%)
large stones	192 / 263 (73%)	55 / 80 (69%)
all	**273 / 350 (78%)**	**289 / 350 (83%)**

- Successful treatments of kidney stones: A open surgery, B a less invasive procedure.
- A is better for small and for large stones, B looks better overall.
- A was given more often to large stones, the harder cases.

Charig CR, Webb DR, Payne SR, Wickham JE (1986). Comparison of treatment of renal calculi by open surgery, percutaneous nephrolithotomy, and extracorporeal shockwave lithotripsy. BMJ 292: 879–882.

Two logistic models, two answers

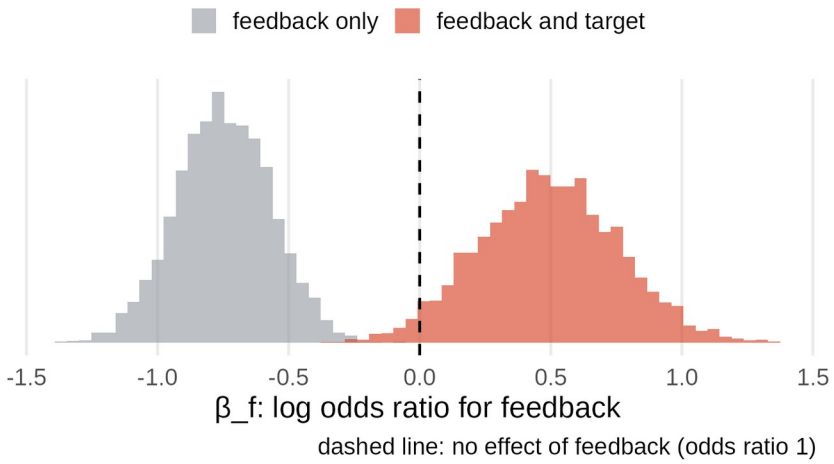
Model 1: $\text{logit}(p_i) = \alpha + \beta_f \cdot f_i$

Model 2: $\text{logit}(p_i) = \alpha + \beta_t \cdot x_i + \beta_f \cdot f_i$

p_i: P(attempt i is a hit); f_i = 1 with feedback, 0 without; x_i = target − 10 (s)

	β_f: log odds ratio, 90%	odds ratio e^β_f, 90%
Model 1	−1.05 – −0.45	0.35 – 0.64
Model 2	0.05 – 0.93	1.06 – 2.53

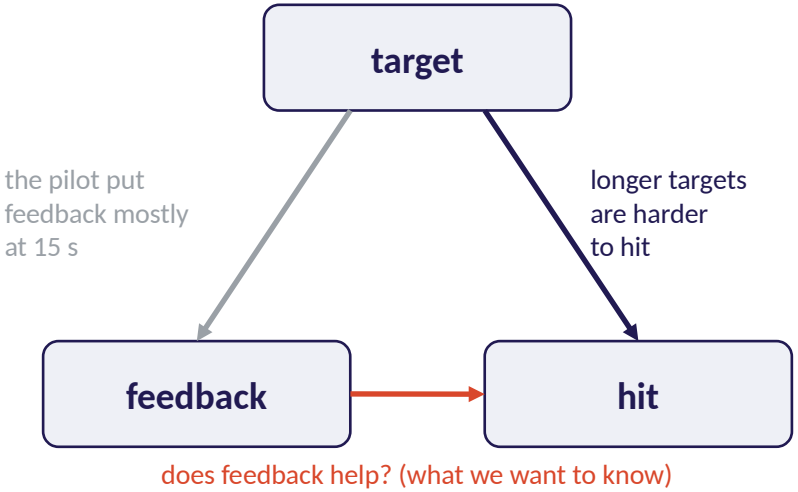
- Model 1 compares attempts with and without feedback, whatever their target: the marginal comparison.
- Model 2 compares them at the same target: the conditional comparison. P(β_f > 0 | y) = 0.97.



Odds ratio = e^β_f

The odds of a hit with feedback ÷ the odds without. Above 1: feedback helps. Model 2: at the same target, feedback multiplies the odds of a hit by 1.06 – 2.53.

Which comparison answers the question?



Arrow: “influences”. The target influences both which condition an attempt is in and whether it is a hit, so it mixes into the overall comparison.

- Question: does feedback help a player at a given target?
- The target influenced both who got feedback (the design) and the hit. Compare at the same target: adjust for it.
- Adjust for what decides whether an attempt gets feedback (here the target), not for what feedback itself changes — for example not for how confident she felt after seeing it.
- The data alone cannot tell you which comparison is right; how the data arose can.
- Randomise when you can: then both comparisons agree, up to chance.

A fair overall comparison

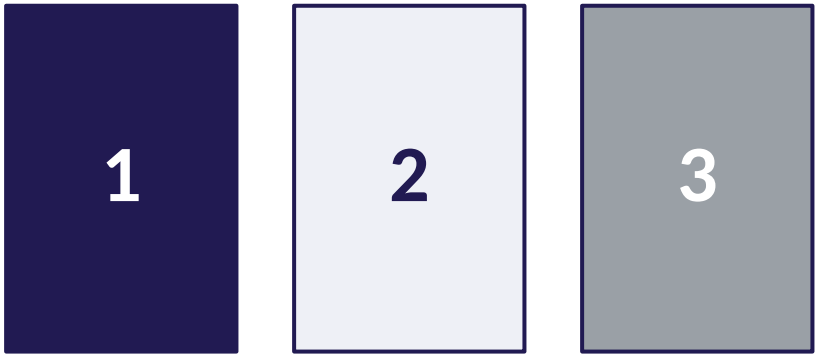
- From Model 2, predict P(hit) at each target, with and without feedback (six rows of X_new).
- Average the predictions over a mix of targets you choose — the same mix for both conditions.

```
p_without <- (draws$`p_new[1]` + draws$`p_new[2]` + draws$`p_new[3]`) / 3   # equal mix
p_with    <- (draws$`p_new[4]` + draws$`p_new[5]` + draws$`p_new[6]`) / 3
quantile(p_with - p_without, c(0.05, 0.95));  mean(p_with > p_without)
```

equal mix of 5, 10 and 15 s	mean	90% interval
without feedback	0.55	0.49 – 0.61
with feedback	0.64	0.59 – 0.69
difference	0.10	0.01 – 0.17

P(difference > 0 | y) = 0.97

The Monty Hall problem



- Behind three doors: one car, two goats. You pick door 1.
- The host knows where the car is. He always opens one of the other doors with a goat behind it — here door 3.
- He offers you a switch to door 2.

Question

Stay with door 1 or switch to door 2? Many people say: two doors left, so 50 : 50.

Bayes' rule, with the host's action as the data

Unknown: where the car is, C = 1, 2 or 3. Data: the host opens door 3.

car behind	prior P(C)	P(host opens 3 | C)	prior × likelihood	posterior
door 1	1/3	1/2 (he picks 2 or 3 at random)	1/6	1/3
door 2	1/3	1 (he must avoid the car)	1/3	2/3
door 3	1/3	0 (he never shows the car)	0	0
sum	1		**1/2**	1

posterior = prior × likelihood / sum: (1/3 × 1) / (1/2) = 2/3 for door 2

Switch: the car is behind door 2 with probability 2/3.

Why switching wins

- Staying wins only if your first pick was right: probability 1/3.
- Switching wins whenever your first pick was wrong: probability 2/3 — the host has removed the other wrong door for you.
- The prior is flat; all the information is in the likelihood: the host's rule depends on where the car is.
- Door 2 gains because the host was forced to leave it closed if the car was there; door 1 does not, because he could open either door if the car was behind it.

Same lesson as Simpson's paradox

The conclusion depends on how the data arose: the host's rule here, the pilot's design there. The numbers alone are not enough.

Change the host's rule

A forgetful host opens door 2 or 3 at random. It happens to be door 3, with a goat.

car behind	prior	P(opens 3 and shows a goat | C)	prior × likelihood	posterior
door 1	1/3	1/2	1/6	1/2
door 2	1/3	1/2	1/6	1/2
door 3	1/3	0 (he would show the car)	0	0

- Exactly the same thing seen — door 3, a goat — but now staying and switching are 50 : 50.
- In Exercise B you check both hosts by simulation, without a template — and then try 100 doors.

Same observation, different data-generating process, different posterior.

The modelling workflow



- Start from the question: which number would answer it, with what uncertainty?
- Write the model as a story of how the data were generated; simulate from it before fitting.
- Check the computation (Day 3), then the model (Days 5–6). They are different checks.
- Revise when a check that matters fails, and say what you changed and why.
- Think about how the data arose before comparing groups (this afternoon).

Bringing it to your own research

Outcome	What kind of number? A measurement, a yes/no, a count? → normal, Bernoulli/binomial, Poisson.
Groups	Are observations grouped (people, sites, batches)? → a hierarchical model.
Predictors	Which ones, on which scale? Centre them so that every parameter means something.
Priors	Priors you can picture, checked by prior predictive simulation.
Checks	Which feature of the data matters for your question? Check that one.

Where to go next

Stan User's Guide and Functions Reference (mc-stan.org/docs)

McElreath: Statistical Rethinking

Gelman, Hill and Vehtari: Regression and Other Stories

Gelman et al.: Bayesian Data Analysis (BDA3)

The final assignment

What	One exercise collection for everybody: seven independent problems, each with its own setting and data
You do	Fill in the blanks in the R code and the Stan templates, run the notebook, report the requested results
You submit	The notebook with outputs (.ipynb), an HTML or PDF export, and a zip of your .stan files, on Moodle
Deadline	13 December 2026, 23:55 (Danish time). Strict: the upload closes then. Upload early, replace later.
Scoring	100 points: 20 each for model formulation, Bayesian reasoning, computation, checking, communication
Rules	Your own work; runs from a fresh kernel; you may be asked to explain any part

You may use AI assistants without declaring it, but you should be able to explain every line.

Passing the course

course score = 0.5 × in-class score + 0.5 × final-assignment score

in-class score	final assignment needed
100%	40
90%	50
80%	60
70%	70
60%	80

- Pass: course score at least 70% and attendance on at least 5 of the 6 days.
- A missed day needs advance notice and a make-up submission.
- Today counts like every other day: work on the exercises and take part in the peer review.

Worked example: Simpson's paradox

Question 1

In the pilot, what would the overall hit rates have been if both conditions had used the same mix of targets as the attempts without feedback (160, 60, 20)? Use the target rates in the table.

Question 2

In the Day 1 games everybody played without feedback first, then with feedback. What else, apart from feedback, differs between the two conditions?

Question 3

Model 2 assumes that the feedback effect on the log-odds is the same at every target. How could you let it differ, and what would the design matrix look like?

Worked example: answers

Question 1

Without feedback: 69% (unchanged). With feedback, at the same weights: (160 × 0.85 + 60 × 0.70 + 20 × 0.39) / 240 = 77%. With equal mixes, feedback is ahead again.

Question 2

Practice and order: with feedback is always the second game, after 30 attempts. A difference may be learning or tiredness, not feedback. Only a design that varies the order (or randomises it) can separate them.

Question 3

Add an interaction column x · f to X: then β for that column is how much the feedback shift changes per second of target. X gets three columns: x, f, x × f.

Exercise B · 13:30 – 15:00

- A pilot study: hits within each target and overall; find the reversal.
- Two logistic models in Stan, and a fair overall comparison.
- The whole class: does feedback help, and does the model fit?
- Challenge, no template: simulate Monty Hall for two hosts, then with 100 doors.
- Your group's question set: concept · computation · pitfall, with worked solutions.

Everyone back at

15:00

peer review B

If Stan does not compile on your laptop: continue on UCloud or a group member’s laptop. New roles this block.

Peer review B · 15:00 – 15:45

20 min

Partner interviews

Round 1: odd group asks even group. Round 2: swap. Two questions each, to named members.

23 min

Plenary

Two groups drawn from those not yet drawn. Then the reporters' hardest open questions.

2 min

Wrap-up

Closing: questions about the final assignment.

Partner pairs: 1 ↔ 2 · 3 ↔ 4 · 5 ↔ 6 · 7 ↔ 8 · 9 ↔ 10

Daily diary · 15:45 – 16:00

menti.com · code: ____ ____

1. Three things I have learned today
2. Three things I am not yet very familiar with
3. What I want to practise before the final assignment